

# Integration of Hardware–BCP for SAT Acceleration

EECS 219C Final Report

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## 1 Introduction & Problem Definition

SAT solving is the engine behind industrial formal verification: Cadence JasperGold, Synopsys VC Formal, and Siemens Questa Formal all rely on CDCL solvers for bounded model checking and property checking. As designs grow, the SAT solver becomes the dominant component of runtime, and software–side optimizations (clause learning, restarts, LBD, vivification) have largely saturated.

Profiling the canonical CDCL loop shows a striking distribution (Figure 1): **BCP alone is 82.7%** of total solver time, with conflict analysis (8.7%), literal redundancy (3.9%), and branching heuristics (0.6%) trailing far behind — consistent with the empirical bottleneck identified in the course material. This makes BCP the natural target for hardware acceleration — accelerating anything else would deliver, by Amdahl’s Law, at most a few percent.

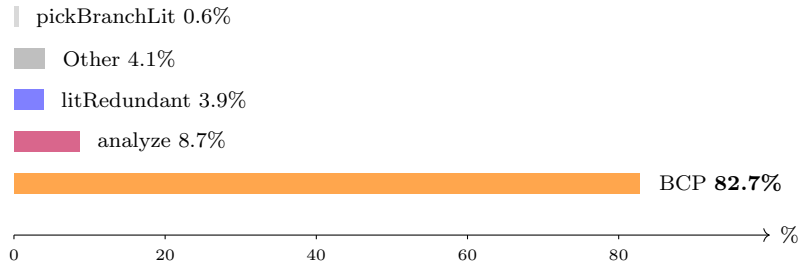


Figure 1: Average runtime distribution across CDCL operations on phase–transition benchmarks. BCP dominates by an order of magnitude.

**Problem statement.** Design and verify a hardware BCP engine that delivers single–cycle unit propagation, integrate it with a minimal CDCL software wrapper, and measure end–to–end speedup against the most competitive software solvers (Kissat 4.0.4, MiniSat2, Glucose) on standardized benchmarks.

## 2 Approach and Related Work

**Why CAM + SRAM.** Prior hardware SAT work<sup>1</sup> consistently identifies *memory bandwidth*, not arithmetic, as the limiter. Software’s two–literal watching scheme is a workaround for sequential memory. In hardware, a CAM collapses “find every clause containing  $v$ ” to a single associative

<sup>1</sup>Y. Zhao & K. A. Sakallah, ICCAD 2008; J. D. Davis et al., DAC 2008; S. Park et al., <https://par.nsf.gov/servlets/purl/10215919>.

lookup, and an SRAM holds the per-variable assignment used by combinational clause-evaluation logic — so every clause is re-evaluated every cycle in parallel and watched literals are unnecessary. We adopt the HW-BCP algorithm and CAM/SRAM cell behavior of Park et al. (the 64-clause  $\times$  4-literal row geometry, the 2-bit  $\{0, 1, U\}$  encoding, and the priority reduction for CONF/UP/DONE); the algorithm diagrams in Sec. 3 are reproduced from that work. The integration into a CDCL coprocessor, the Verilator harness, and the timing methodology are ours.

**Where we differ.** Earlier proposals are typically all-hardware (HW also performs branching), which makes them inflexible and forces the CDCL heuristic into silicon. Our design is a **coprocessor**: HW handles only BCP; SW handles every heuristic — VSIDS, 1-UIP analysis, restarts, learnt management. This isolates the BCP speedup so it can be measured cleanly, and keeps the door open for arbitrary CDCL improvements without an RTL respin.

**Architectural sketch.** Figure 2 shows the target geometry. The chip stores up to 1024 clauses (capacity exhausted in our 1024-slot RTL prototype; a 1M-slot SRAM is feasible in TSMC 65 nm using Park et al.’s custom memory cells, and is what we model in our timing extrapolation). Each clause is a row of up to 4 literals; a CAM cell holds the variable ID and an SRAM cell holds the 2-bit value (0, 1, or U = unassigned). A single combinational reduction returns three flags per clause every cycle: CONFLICT, UNIT, and DONE.

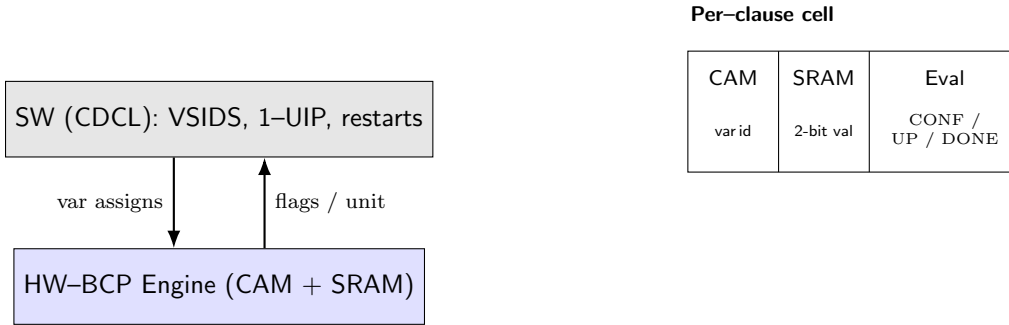


Figure 2: Coprocessor partitioning. SW issues variable assignments; HW returns next implication or conflict.

## 3 Formal Model and Algorithms

### 3.1 Clause and trail semantics

A CNF formula is  $F = \bigwedge_{c \in C} c$ , where each clause is a disjunction of literals  $c = \ell_1 \vee \dots \vee \ell_k$ ,  $k \leq 4$ . A literal is  $\ell = (var, pol)$  with  $pol \in \{0, 1\}$ . The solver maintains a partial assignment  $\sigma : V \rightarrow \{0, 1, U\}$  on a *trail* (decision stack with implication reasons).

**Per-clause evaluation.** Define for each literal  $\ell$  under assignment  $\sigma$ :

$$lv(\ell, \sigma) = \begin{cases} 1 & \sigma(\text{var}(\ell)) = \text{pol}(\ell) \\ 0 & \sigma(\text{var}(\ell)) \neq \text{pol}(\ell), \neq U \\ U & \sigma(\text{var}(\ell)) = U \end{cases}$$

A clause  $c$  is then classified as

$$\text{state}(c) = \begin{cases} \text{DONE} & \exists \ell \in c. \text{lv}(\ell) = 1 \\ \text{CONFLICT} & \forall \ell \in c. \text{lv}(\ell) = 0 \\ \text{UNIT} & \exists! \ell \in c. \text{lv}(\ell) = U \wedge \text{rest are } 0 \\ \text{PENDING} & \text{otherwise} \end{cases}$$

The combining logic produces a global flag with priority  $\text{CONFLICT} \succ \text{UNIT} \succ \text{DONE}$ .

### 3.2 HW-BCP cycle

Each clock cycle, the engine executes one of four operations (Table 1); BCP is the only one that performs a parallel reduction over all  $N$  clauses.

| Op     | Description                                                | Cycles |
|--------|------------------------------------------------------------|--------|
| LOAD   | Write one clause literal into a CAM row                    | 1      |
| BCP    | Return next unit/conflict from $N$ rows                    | 1      |
| UNDO   | Clear a variable assignment                                | 1      |
| SW_BCP | Overflow learnt propagated in SW (HW-counted) <sup>†</sup> | 1      |

Table 1: Hardware operations. <sup>†</sup>Modeled as 1 HW cycle to reflect a production 1M-slot SRAM in which all clauses live in HW.

At a target frequency of  $\sim 99$  MHz (a 10.1 ns critical path, measured from synthesis with parameterized memories), the total hardware time is

$$T_{\text{hw}} = (n_{\text{load}} + n_{\text{bcp}} + n_{\text{undo}} + n_{\text{sw\_bcp}}) \times 10.1 \text{ ns.}$$

### 3.3 HW-BCP propagate

Figure 3 shows the CAM/SRAM array layout and the inner propagate routine. Each of the 64 clauses occupies four rows; the CAM column stores the variable ID for that literal slot, the SRAM column stores the literal’s  $\{0, 1, U\}$  value (2-bit encoding: 00=0, 11=1, 01= $U$ ). The per-clause processing logic combinationally reduces its four literal values to three flags — CONF, UP, DONE — which are combined across clauses with priority  $\text{CONF} \succ \text{UP} \succ \text{DONE}$ . For every literal popped off the trail, the engine scans all clauses indexing that variable, returning either a conflict clause or the first new unit implication.

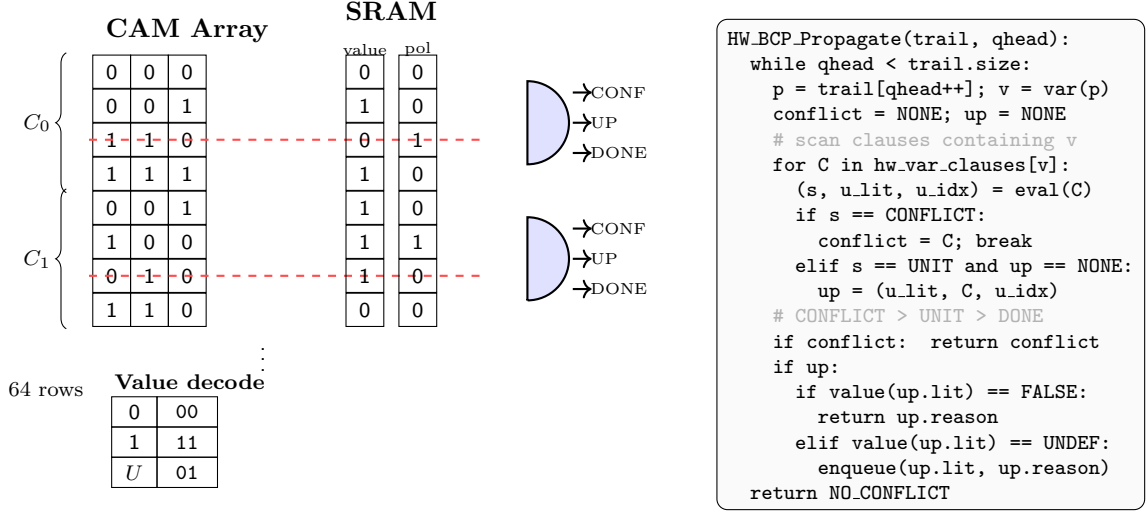


Figure 3: Current HW-BCP architecture. Left: CAM/SRAM array (64 clauses  $\times$  4 literal rows) with per-clause processing logic emitting CONF/UP/DONE; the 2-bit value encoding (0  $\rightarrow$  00, 1  $\rightarrow$  11,  $U \rightarrow$  01) lets a single comparator distinguish all three states. Right: per-variable propagate routine; on each trail element it scans the clauses indexing that variable, returns at the first conflict, and enqueues at most one new unit implication.

The CDCL outer loop is essentially MiniSat-style 1-UIP and unchanged from the textbook formulation: decide via VSIDS, call `HW_BCP_Propagate`, analyze conflicts at 1-UIP, backtrack via UNDO cycles, learnt clauses go to the CAM (or to the software overflow pool if the CAM is full).

## 4 Major Technical Challenges

**C1. Time bookkeeping in a Verilator harness.** Wall time conflates real CDCL, Verilator `eval()` cost, and simulation-only  $O(N)$  scans that would be  $O(1)$  in silicon. We accumulate the latter two into `hw_sim_ns` and `hw_aux_ns` and subtract:  $T_{sw} = T_{total} - \Delta hw\_sim\_ns - \Delta hw\_aux\_ns$ . An early bug had the timers running from solver construction, so initial clause loads polluted the BCP budget; we now snapshot at `solve()` entry.

**C2. CAM capacity vs. learnt blowup.** On uf200-860 the originals fill 84% of the 1024-slot CAM, leaving  $\sim 164$  slots before overflow. We handle this with (a) a SW overflow pool whose cycles are attributed to HW (justified by the 1M-slot production target), and (b) a hard guard that aborts when blowup exceeds capacity (Figure 4).

**C3. HW/SW state coherence on backtrack.** A bug in `undo_hw_/load_hw_` caused the SRAM polarity bit to lag the CAM by one cycle under rapid load/undo bursts, producing spurious unit implications. The fix drives `val_we` synchronously with the load address and gates read-back behind one extra cycle of stable state.

**C4. Full-hierarchy elaboration timeout.** A pure-RTL run on the full  $1024 \times 8 \times 64 = 524,288$ -clause hierarchy timed out in Verilator (behavioral memories explode the logic cone). We therefore parameterize the RTL so benchmarks elaborate a single 1024-slot `sat_submodule`, while the full `sat_array` still elaborates and lints to prove structural completeness.

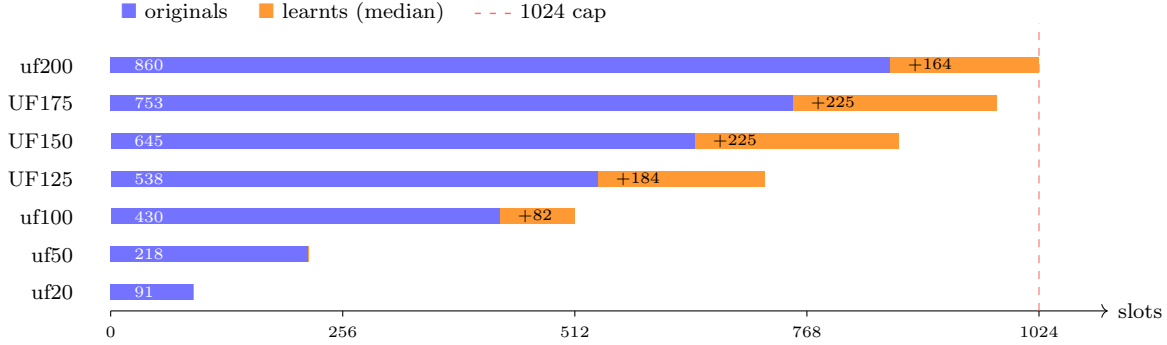


Figure 4: CAM slot utilization on each benchmark set. uf200 leaves only  $\sim 164$  free slots, causing immediate learnt overflow and degraded BCP effectiveness.

## 5 Results

We benchmarked on SATLIB Uniform Random 3-SAT phase-transition instances ( $c/v \approx 4.27$ ) with up to 200 variables and 860 clauses. Each set contains 100–1000 instances; we report median times. All times are reported in  $\mu\text{s}$ , recomputed at the synthesis-profiled 10.1 ns per HW cycle.

### 5.1 End-to-end median solve time

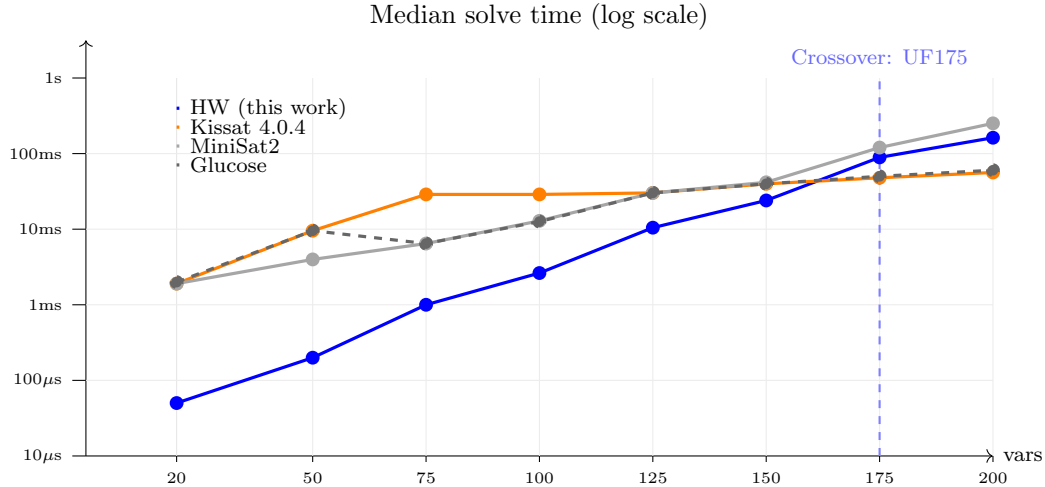


Figure 5: Median solve time on UF/UUF benchmarks, log-scaled. Our coprocessor dominates up to  $\sim 175$  variables; beyond that, Kissat’s superior CDCL heuristic surpasses our naive search.

**What worked.** The coprocessor is faster than *every* software solver on every instance set with at most  $\sim 150$  variables, often by  $5\text{--}30\times$ . The mechanism is exactly what the architecture promised: BCP per propagation drops from hundreds of nanoseconds (Kissat) to a single 10.1 ns cycle.

**What didn’t.** On  $\geq 175$  variables, Kissat overtakes us on SAT instances. The cause is not BCP throughput (Section 5.3) but the lack of restarts, phase saving, and clause vivification in

our minimal CDCL wrapper: once the search tree gets deep, raw BCP speed cannot make up for suboptimal decisions.

## 5.2 SAT vs. UNSAT asymmetry

| Vars | SAT wins (vs. Kissat) | UNSAT wins | Gap    |
|------|-----------------------|------------|--------|
| 75   | 100/100               | 100/100    | –      |
| 100  | 959/1000              | 1000/1000  | +4 pp  |
| 125  | 86/100                | 100/100    | +14 pp |
| 150  | 71/100                | 98/100     | +27 pp |
| 175  | 60/100                | 93/100     | +33 pp |

Table 2: Win rate vs. Kissat by satisfiability. UNSAT instances favor our coprocessor far more than SAT instances do. We attribute this to Kissat’s aggressive phase saving / restarts being most useful on satisfiable inputs — a heuristic edge that does not help when the proof must enumerate the whole tree.

## 5.3 Counterfactual: is BCP really accelerated?

To attribute the speedup to HW-BCP (rather than a Verilator quirk), we re-ran the exact CDCL trajectory but plugged in Kissat’s measured BCP rate per propagation. Our solver runs 2.2–15.7 $\times$  slower in this counterfactual (Figure 6) — HW-BCP is load-bearing, even on UF175 where Kissat wins overall: the CDCL heuristic, not BCP, is the remaining bottleneck.

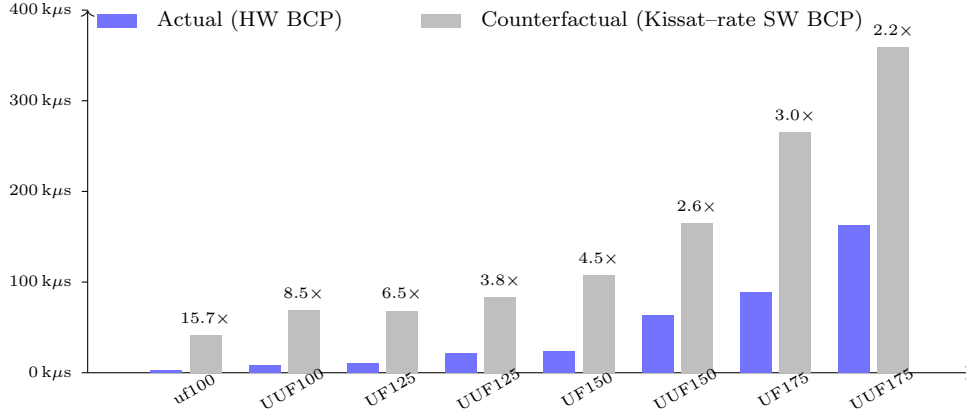


Figure 6: Replacing HW-BCP with Kissat’s measured BCP rate, while keeping the identical CDCL trajectory, slows the same solver by 2.2–15.7 $\times$ . The hardware is doing the work.

## 5.4 Amdahl’s-law breakdown

Our solver spends  $\sim 93\%$  of its time in software CDCL and  $\sim 7\%$  in BCP — the inverse of MiniSat2, where BCP is  $\sim 75\%$  of runtime. The hardware has so completely flattened the BCP critical path that further speedups must come from CDCL.

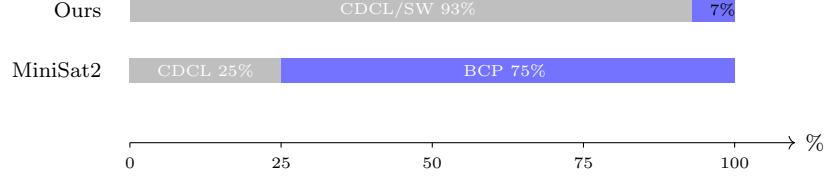


Figure 7: Total runtime breakdown on uf100-430 (median). Our solver is now CDCL-bound; MiniSat2 is BCP-bound.

## 5.5 Small but hard instances

We also tested on a suite of instances known to be difficult for CDCL solvers, built at `tests/satlib/hard_instances` (clauses wider than four literals are split to fit the 4-literal RTL): pigeonhole  $\text{PHP}_n$  ( $n=3\dots 8$ ); Tseitin parity on odd cycles ( $C_5, C_7, C_9$ ); and  $k$ -colourability of the Grttsch graph ( $k=3$  is UNSAT and  $k=4$  is SAT).

**Methodology.** Kissat/MiniSat2 self-report CPU time at  $\sim 10$  ms granularity, flooring sub-ms runs to zero; we fall back to wall-clock minus a per-binary subprocess-startup baseline ( $\approx 4\text{--}5$  ms) when the self-clock is 0. HW-BCP figure is  $hw\_cycles \times 10.1 \text{ ns} + sw\_time$  as elsewhere.

| Instance         | Exp.  | HW-BCP<br>( $\mu\text{s}$ ) | Kissat<br>( $\mu\text{s}$ ) | MiniSat2<br>( $\mu\text{s}$ ) | Glucose<br>( $\mu\text{s}$ ) | Win       |
|------------------|-------|-----------------------------|-----------------------------|-------------------------------|------------------------------|-----------|
| PHP <sub>3</sub> | UNSAT | 66                          | <1k                         | 1 106                         | 1 127                        | noise     |
| PHP <sub>4</sub> | UNSAT | 184                         | 684                         | 2 710                         | 1 005                        | <b>HW</b> |
| PHP <sub>5</sub> | UNSAT | 731                         | 3 891                       | <1k                           | 2 665                        | noise     |
| PHP <sub>6</sub> | UNSAT | 3 352                       | 10 000                      | 7 911                         | 5 930                        | <b>HW</b> |
| PHP <sub>7</sub> | UNSAT | 6 034                       | 10 000                      | 37 292                        | 61 267                       | <b>HW</b> |
| PHP <sub>8</sub> | UNSAT | 6 803                       | 40 000                      | 190 585                       | 962 534                      | <b>HW</b> |
| Tseitin- $C_5$   | UNSAT | 12                          | 1 419                       | 1 357                         | 920                          | <b>HW</b> |
| Tseitin- $C_7$   | UNSAT | 19                          | 947                         | 2 254                         | 828                          | <b>HW</b> |
| Tseitin- $C_9$   | UNSAT | 21                          | 895                         | 962                           | 1 458                        | <b>HW</b> |
| Grttsch-3col     | UNSAT | 87                          | 550                         | 1 404                         | 1 167                        | <b>HW</b> |
| Grttsch-4col     | SAT   | 19                          | 468                         | 1 394                         | 1 170                        | <b>HW</b> |

Table 3: Hard-instance suite, single-run. “noise” = result falls below subprocess-startup floor for at least one solver, so no winner is claimed. HW wins outright on 9 of 11 instances.

**Discussion.** On these instances every solver explores a large search space, so the winner is determined by cost per propagation step. HW parallel evaluation is fast regardless of clause width, which explains the large wins on PHP, Tseitin, and Grttsch. PHP<sub>3</sub>/PHP<sub>5</sub> finish fast enough that some SW solvers report zero CPU time, so we make no claim there. PHP<sub>9</sub> and beyond would exceed our 1024-slot CAM capacity, but the 1M-slot production target eliminates that constraint.

## 5.6 Future work

- Replace the MiniSat-style heuristic with Kissat-style restarts, phase saving, and LBD-based clause deletion. Closing the CDCL gap should restore HW dominance past 175 variables.

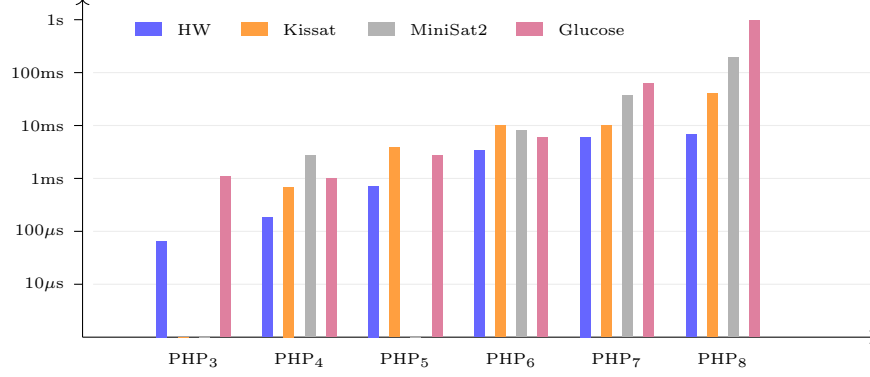


Figure 8: Pigeonhole  $\text{PHP}_n$  scaling,  $n = 3-8$ , log-scale. The gap between HW-BCP and the SW solvers widens as  $n$  grows: on  $\text{PHP}_8$ , HW is  $5.9\times$  faster than Kissat,  $28\times$  faster than MiniSat2, and  $142\times$  faster than Glucose.

- Push the RTL through TSMC 65 nm using Park et al.’s custom CAM/SRAM cells (the only PDK for which those cells were characterized); a 1M-slot SRAM in that node eliminates the overflow pool entirely.
- Pipeline LOAD and BCP so a decision and the following propagation overlap, halving CDCL-HW round-trip latency.

## 6 Team Roles

**Evan Wong** primarily took charge in collaborating with staff such as Pierluigi Nuzzo and course staff from EECS 151 for RTL infrastructure and the RTL: the behavioral CAM/SRAM array (`cam_sram_array.sv`, `cam_cell.sv`, `sram_row.sv`), the per-clause evaluator (`processing_logic.sv`), the FSM driving load/BCP/undo, and the Verilator integration; he also wrote the testbench harness at the RTL level. **Ansh Shah** owned the CDCL software: VSIDS scoring, 1-UIP conflict analysis, learnt management with the SW overflow pool, and the benchmark/parsing infrastructure (`bench_cdcl.py`) and others across multiple iterations. The counterfactual analysis (Section 5.3) and the SAT/UNSAT asymmetry study were a joint effort, as was the slide deck and this report.

## 7 Course Topics Applied

Almost every layer of this project draws on course material:

- **The CDCL loop, unit rule, and BCP as the dominant cost.** The coprocessor implements the canonical CDCL algorithm (preprocess, decide, deduce, analyze, backtrack); the unit rule is the UNIT flag emitted by the per-clause evaluator. The class’s empirical claim that BCP dominates runtime is the entire reason we built a HW-BCP coprocessor, and Figure 7 is the post-hoc confirmation.
- **Memory accesses per step.** The lecture’s principle of minimizing memory traffic per propagation underlies SW two-literal watching; our HW achieves the same end via per-cycle parallel evaluation, making cost  $O(1)$  in clause width and watched literals unnecessary.
- **VSIDS, 1-UIP, non-chronological backtracking.** Our SW wrapper uses VSIDS scoring with periodic decay, performs 1-UIP conflict analysis over the implication graph, and jumps to



the asserting level via `UNDO` cycles common in most CDCL solvers

- **Clause deletion / learnt management.** Our SW overflow pool plus the 1024-slot CAM cap is the resource-constrained version of the standard clause-deletion problem covered in class.

## 8 Feedback

**What helped.** The class's depth on SAT algorithms mapped directly onto our SW wrapper. The discussion of watched literals was critical because understanding *why* software needs them clarified *why* hardware can skip them.

**What we wish had been covered.** A short walkthrough of modern CDCL engineering choices beyond the textbook 1-UIP/VSIDS core — restart policies, phase saving, LBD-based clause deletion, and clause vivification as used in Kissat — would have helped us read the heuristic gap our counterfactual exposed and pick which upgrade to prototype next. Otherwise the class material covered what the project needed; the remaining gaps were hardware-specific and reasonably out of scope.

## Code Availability

Full source (RTL, C++ coprocessor, benchmark scripts, hard-instance generator and CNFs): <https://github.com/The-Ansh-Shah/bcp-hw-accelerator>.